

The verdict, on one page

THE QUESTION

Does a model of a compressor's physics, calibrated on one healthy month and never shown a failure, warn about air leaks better than the alarm the machine already carries?

THE METHOD, IN FIVE LINES

1. **1,516,948 readings** of 15 signals, one every ten seconds, seven months of the air compressor of a Porto Metro train: 208.2 MB of CSV, from the UCI, under a CC BY 4.0 licence.
2. A **layered data lake**, with the readings laid on a regular ten second grid, and the same readings served from PostgreSQL too.
3. **Four numbers measured in February** describe the machine. It starts at 8.05 bar, stops at 10.10, fills at 1.1796 bar per minute and the plant draws 0.0711. The air going in and the air going out balance to 0.7 %.
4. The **twin** runs alongside with those four numbers frozen. The only thing anyone looks at is its disagreement with the machine, the **residual**, in bar per minute.
5. The threshold is not chosen: it is **swept whole**. The highest one that still catches the four documented failures is 1.15 bar per minute.

THE VERDICT

OVER THE 207 DAYS THE TWIN CAN JUDGE	THE TWIN	THE ALARM FITTED
Failures detected	4 of 4	2 of 4
Days with an alarm	18	94
False alarms a month	1.5	11.4

What it does not do: predict. Only one of the four warns beforehand, and by a single day. An air leak does not grow slowly: it opens, and the compressor goes from working a twentieth of the time to never stopping in a few hours.

What it found without being asked. Twelve days of March with the machine failing and no maintenance report, and four independent signals say so at once. Counting March as a failure, the result is 5 of 5 at 1.2 false alarms a month.

WHAT WOULD BREAK IN A REAL PLANT

- **One machine only.** The four numbers belong to this compressor; another one needs them measured again, and one healthy month is enough.
- **Four failures are not a sample.** This is a study of four cases, and one hit more or less moves the result.
- **Without the receiver's volume there are no litres.** The residual is in bar per minute, and turning it into a flow rate needs a number the datasheet does not give.
- **No reference leak.** Nobody opened a valve of known size, so the scale of severity is not calibrated.
- **The machine drifts.** Healthy demand doubles between February and August, and the frozen threshold marks 24 days of August out of 31: it needs recalibrating, or reading each day against the fourteen before it.
- **The failure log is data with holes too,** and the model is judged against it.
- **Detecting is not warning:** the threshold that gets 4 of 4 fires with the compressor almost flat out, so it states a disaster already under way.
- **A portable PostgreSQL is not a production database:** no automatic backups, no users, and nothing holding it up if it falls over.

HOW TO CHECK IT

Every number on this page comes out of running a script, and ten checks stand between a lesson and a commit. The course is 31 modules, the panel shows the 214 days, and the code is all at github.com/kmortizva-data/fuelle.